

Nanotubes

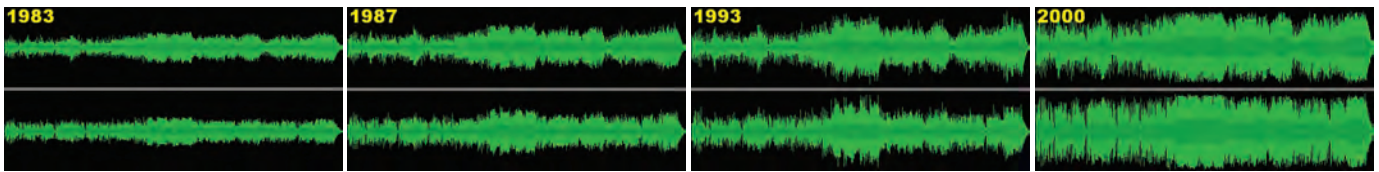
Like so much else, Carbon Nanotubes may end up in speakers of the future, letting even laptop speakers dish out loud, concert-like music



WebEar Headphones

Andy Kurovets and Kris Dekeijser have designed a futuristic headphone that doubles up as a webcam

From the labs



The sound wave for “Something”, by The Beatles, from its original(ish) form in 1983, to what it became by 2000. You’ll notice that not only has the song been made much louder, but it’s been boosted disproportionately. The original has a few quiet and a few not-so-quiet bits, which have now been turned into annoyingly loud or annoyingly slightly-less-loud

the wall. And the big selling point here is that this works even with regular 5.1 audio as well, so it even makes business sense. Right. Here we are, then.

Interlude

We don’t normally do this, but permit us, if you will, to get up on our soapbox and yell for a bit. You see, as the beginning of this article plainly suggests, the “future” of technology already exists. With lossless formats like FLAC, and gigabytes of space available for peanuts, it’s entirely possible for us to enjoy music exactly as the artists recorded it. You’ll have to part with many months worth of family income, but it’s there. We don’t get to enjoy this because of good old human idiocy.

Here’s how we got the idea for a “future of sound” article: we’ve gone from CRT monitors to 22-inch HD capable screens in almost no time. We’ve gone from lugging around CDs and a portable CD player in a backpack, to

dumping entire music collections on to a PMP and still have place left over. Our experiences with audio, on the other hand, haven’t changed in decades. Whether today or in 1979, there’s only one way to bring a “concert experience” home — get the music in its purest form (no, not by kidnapping the band), put it through the best equipment money can buy (ideally the kind that the studios own), and turn up the volume. With IOSONO, our concert experience may come closer to the experience of being in the studio with the artist. And if you can set up an IOSONO system at home, you can bring that experience home as well.

But the real problem — the rotten banana skin that even the most agile technology will slip on — is what’s known as the “loudness war”. At some point in the jukebox era, recording industry executives decided that consumers were such brain-dead twits that they couldn’t be asked to get up and turn up the volume when the noise

in the bar/café/establishment-of-your choice was drowning out the song. And so they decided to make “hot” masters — recorded audio whose volume is artificially boosted to make them stand out more when they’re played. This came to be regarded as a really good idea, so they kept doing it. We’ve laboured under the impression that the audio on CDs is as good as it gets, but in reality, they’re shining examples of artificially loud music — the boosted track requires less digital space, and so it made sense to use this technique to cram in as many songs as possible.

The worst part is that even though the audio CD, for all practical purposes, is dead, this practice continues. So even though you can get yourself the best equipment, the old garbage-in-garbage-out rule applies, and all that money would be spent for naught.

But does this matter? When we take into account economics, the demands of consumers, and other such factors that have nothing to do with technology, we come to one key phrase...

Good enough

If you, like so many of us, like your music in your pocket while you travel, do you really think — even with sound-cancelling headphones — that you’ve ever cared that the music is artificially “enhanced”? Or do you just kick back and enjoy? A sound-gasm may be highly desirable, but its lack isn’t really going to tear the world in two. If you can get joy out of sound without caring what bitrate it’s reaching you at, then you probably aren’t even bothered about what this article is saying.

If we demand great technology, it may or may not come. At least there’ll always be great music. **d**

THE AUDIO SPOTLIGHT

Since we have already touched on the subject of performance audio, we should also mention another form of audio in public places — obnoxious public advertising. If you’ve been in a mall or supermarket and heard the repeated wail of an announcement or advertisement you didn’t really care about, you’ll be interested to know that it’s possible they’ll be obsolete soon.

Like most audio innovations, the “audio spotlight” has been around for ages — in fact, we read about it in Digit way back when we were readers ourselves.



An audio spotlight prototype by the Kakamura Laboratory at the University of Electro-Communications, Japan

The basic principle is this: instead of transmitting audible sound waves, transmit ultrasound waves that will degenerate in the air in such a way that they turn into regular sound waves by the time they reach your ears. This degeneration can be controlled quite precisely, too.

Ultrasonic waves are directional — they don’t get transmitted everywhere, just in the direction the speaker is pointing in. This means

that you could sit right in front of your TV and watch your soaps on full blast, while the rest of your family could sit as little as a foot to your right and not hear a thing, leaving them free to think about what a disappointment you’ve become.